

M003 Extension — English Worksheet (Intermediate)

Topic: FizzBuzz Pixel Art (remainders & rules) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Week 3) · **Time:** about 40 minutes

You already made pixel art by choosing each block's colour **by hand**. This time the **code** chooses every colour for you — using one rule, the **FizzBuzz rule**. Load the world, type **run** in the chat, and watch the agent paint a **15 × 15** wall one block at a time, counting from **1 to 225**.

If the block's number...	it says	colour
divides by 3 and 5	FIZZ BUZZ	● gold
divides by 3	FIZZ	■ black
divides by 5	BUZZ	■ purple
divides by neither	<i>(nothing)</i>	■ green

Big idea: a block's colour comes from its **number**. The code divides the number by 3 and by 5 and looks at the **remainder** (what's left over). Remainder **0** means it divides evenly.

Here is what one row (blocks 1 → 15) looks like. This same row repeats all the way up the wall.

🔑 **One row = blocks 1 → 15 (this row repeats up the whole wall)**

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

1 · Predict 🌟

Use the rule. Write each block's colour and a short reason.

Block 6 → (divides by __ , so colour __)

Block 5 →

Block 15 →

Block 8 →

Block 15 divides by both 3 and 5. Which colour wins — gold, black, or purple? Circle one: gold · black · purple

2 · Spot the Bug 🐛

Bug A — the order is upside down. Gold should win when a number divides by 3 **and** 5. But the $\div 3$ check runs first and grabs block 15 before the gold check.

```
if remainder of  $N \div 3 = 0$ :  
    paint BLACK    ← FIZZ  
else if remainder of  $N \div 3 = 0$  and remainder of  $N \div 5 = 0$ :  
    paint GOLD     ← FIZZ BUZZ
```

What colour does block 15 wrongly get? Circle one: gold · black · purple

Which check must go first to fix it? Circle one: the $\div 3$ check · the $\div 3$ **and** $\div 5$ check

Bug B — wrong divisor. The purple **BUZZ** stripe should mark blocks that divide by **5**, but this line uses **4**.

```
else if remainder of  $N \div 4 = 0$ :  
    paint PURPLE    ← BUZZ
```

Fill the blank to fix it:

```
else if remainder of  $N \div \underline{\hspace{1cm}} = 0$ :  
    paint PURPLE    ← BUZZ
```

Which number fills the blank? Circle one: 4 · 5

3 · Show Your Work 📷 🎥

Load the world. Type run in the chat and watch the agent count **1** → **225**, painting each block by the rule.

 **The finished wall — 15 × 15 (block 1 → 225)**

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
14	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
13	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
12	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
11	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
10	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
9	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
8	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
7	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
6	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
5	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
4	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
3	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
2	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow
1	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	yellow

The wall is 15 rows, all the same, so you see vertical stripes.

Which columns are black (FIZZ)? Write the numbers:

Modify challenge. Change **one** thing — swap a colour, or make gold mark ÷ **2** and ÷ **3**. Write what you changed, then build it.

Record **one video** (a phone is fine). Show two things:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the colours and the stripes.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.